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 Studierichting: Informatica
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$$\begin{aligned}
 & \int \frac{x - \sin x}{2 \sin^2 \left(\frac{x}{2} \right)} dx \\
 &= \frac{1}{2} \int \frac{x}{\sin^2 \left(\frac{x}{2} \right)} dx - \frac{1}{2} \int \frac{\sin x}{\sin^2 \left(\frac{x}{2} \right)} dx \\
 & \left\{ \begin{array}{l} u = x \\ dv = \frac{dx}{\sin^2 \left(\frac{x}{2} \right)} \end{array} \Rightarrow \left\{ \begin{array}{l} du = dx \\ v = -2 \cot \left(\frac{x}{2} \right) \end{array} \right. \right. \\
 &= \frac{1}{2} \left(-2x \cot \left(\frac{x}{2} \right) + 2 \int \cot \left(\frac{x}{2} \right) dx \right) - \frac{1}{2} \int \frac{\sin x}{\sin^2 \left(\frac{x}{2} \right)} dx \\
 &= -x \cot \left(\frac{x}{2} \right) + \int \cot \left(\frac{x}{2} \right) dx - \frac{1}{2} \int \frac{2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)}{\sin^2 \left(\frac{x}{2} \right)} dx \\
 &= -x \cot \left(\frac{x}{2} \right) + \int \cot \left(\frac{x}{2} \right) dx - \int \frac{\cos \left(\frac{x}{2} \right)}{\sin \left(\frac{x}{2} \right)} dx \\
 &= -x \cot \left(\frac{x}{2} \right) + \int \cot \left(\frac{x}{2} \right) dx - \int \cot \left(\frac{x}{2} \right) dx \\
 &= -x \cot \left(\frac{x}{2} \right) + C
 \end{aligned}$$

References